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ROYAL UNIVERSITY OF PHNOM PENH

បុគ្គលិកបទប្បញ្ញត្តិការណ៍ស្រុក
High-Availability Campus Network Design & Implementation: Secure
Layer 3 Redundancy using HSRP and Inter-VLAN Routing

A Research Report
In Partial Fulfilment of the Requirement for the Degree of
Master of Science in Information Technology Engineering

Seng Heng
Tong Borey

December 2025

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ABSTRACT

In the modern digital era, the reliability and security of campus networks are fundamental to the operational continuity of educational institutions. Traditional legacy network architectures often suffer from "flat" topologies, characterized by single points of failure, inefficient broadcast domains, and a lack of granular security controls. This research project presents the design, implementation, and simulation of a **High-Availability Campus Network** aimed at eliminating network downtime and mitigating internal security threats.

The study employs a hierarchical "Collapsed Core" network model, simulated within the **Cisco Packet Tracer** environment. The core technical strategy involves the deployment of the **Hot Standby Router Protocol (HSRP)** to provide seamless default gateway redundancy. This ensures that a hardware failure at the primary core switch triggers an automatic, transparent failover to a standby device without disrupting end-user connectivity. Furthermore, network segmentation is achieved through **VLANs (Virtual Local Area Networks)** and **Inter-VLAN Routing**, isolating Student, Staff, and Management traffic to optimize bandwidth and enhance privacy.

To address scalability and management efficiency, a **Split-Scope DHCP** architecture is implemented, distributing IP address allocation across redundant core devices to prevent service outages. Security is further hardened at the access layer through **Port Security** with sticky MAC address learning, and at the management plane through **SSHv2** encryption.

Simulation results validate that the proposed architecture achieves 100% uptime during core link failures and effectively blocks unauthorized hardware connections at the network edge. This research concludes that implementing Layer 3 redundancy and logical segmentation provides a cost-effective, enterprise-grade solution for stabilizing university network infrastructures.

Keywords: *HSRP, Campus Network Design, Network Redundancy, Inter-VLAN Routing, Port Security, Split-Scope DHCP.*

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CHAPTER 1: INTRODUCTION

1.1 Background to the Study

In the modern academic era, information technology infrastructure serves as the backbone of educational institutions. The reliability, scalability, and security of a campus network are no longer optional luxuries but fundamental requirements for operational continuity. University networks differ significantly from standard enterprise networks; they must support a highly transient user base (students), secure sensitive administrative data (staff), and maintain high availability for critical research activities.

Traditionally, campus networks were designed with "flat" architectures, relying on basic connectivity without logical segmentation. However, as the number of connected devices—ranging from laboratory workstations to personal laptops and mobile devices—grows exponentially, these legacy designs face critical bottlenecks. Issues such as broadcast storms, IP address conflicts, and security vulnerabilities caused by unauthorized access have become prevalent. Furthermore, the increasing reliance on digital resources demands a network architecture that guarantees uptime; a single router failure should not disrupt the connectivity of an entire faculty.

This research project proposes the design and simulation of a robust, hierarchical campus network. By leveraging advanced switching and routing technologies—specifically VLAN segmentation, First Hop Redundancy Protocols (HSRP), and automated address management (DHCP)—the study aims to create a blueprint for a network that is both resilient to hardware failure and secure against internal threats.

A	B	C
Year	Avg Daily Traffic (GB)	Notes
2014	120	Legacy Network
2015	150	
2016	210	Introduction of BYOD
2017	280	
2018	350	Video Streaming adoption
2019	450	
2020	800	COVID-19 Spike (Remote Learning)
2021	950	Hybrid Learning
2022	1,100	4K Video & Zoom
2023	1,350	Cloud Apps
2024	1,600	Current Demand

Figure 1.1: Evolution of Campus Network Traffic (Source: Estimated data based on global industry trends and Cisco VNI reports)

1.2 Problem Statement

Despite the availability of advanced networking protocols, many educational institutions struggle with network architectures that are fragile and insecure. Specifically, the following technical deficiencies are often observed in legacy network designs:

1. **Single Point of Failure:** In standard designs, a single Core Router acts as the default gateway for the entire campus. If this device suffers a hardware failure or power outage, the entire network loses internet connectivity, halting all administrative and academic operations.
2. **Inefficient IP Management:** Manual assignment of IP addresses (Static IP) or non-redundant DHCP services leads to administrative overhead and frequent IP conflicts, particularly when students connect personal devices to the network.
3. **Security Vulnerabilities:** Flat networks allow students (VLAN 10) to potentially access sensitive Staff resources (VLAN 20) via Layer 2 broadcasting. Additionally, unused physical ports on switches often remain active, leaving the network vulnerable to "rogue device" attacks where an unauthorized user can simply plug in a cable and gain access.
4. **Lack of Hierarchy:** Without a structured approach (Core vs. Access layers), troubleshooting becomes complex, and broadcast traffic degrades overall network performance.

1.3 Aim and Objectives of the Study

The primary aim of this study is to design, simulate, and validate a High-Availability Campus Network Architecture that mitigates the risks of downtime and unauthorized access.

To achieve this aim, the following specific objectives have been established:

- To design a **Hierarchical Network Topology** separating the Core Layer (Routing/Redundancy) from the Access Layer (User Connectivity).
- To implement **VLAN Segmentation** to logically isolate Student, Staff, and Management traffic, enhancing security and reducing broadcast domains.
- To deploy the **Hot Standby Router Protocol (HSRP)** to ensure gateway redundancy, allowing the network to self-heal instantly in the event of a Core Switch failure.

- To configure a **Split-Scope DHCP System** across multiple Core devices to prevent IP address exhaustion and ensure high availability of IP allocation services.
- To implement **Layer 2 Security measures**, specifically Port Security and Sticky MAC addresses, to prevent unauthorized hardware connections.

1.4 Rationale of the Study

The rationale for this research lies in the critical need for "Always-On" connectivity in education. For network engineers and IT administrators, understanding how to transition from a basic network to a redundant, enterprise-grade architecture is a vital skill.

By simulating this environment, the study provides a proof-of-concept for cost-effective network upgrades. It demonstrates that reliability can be achieved through logical configuration (HSRP, VLANs) rather than simply purchasing expensive hardware. Furthermore, the focus on security (SSH, Port Security) addresses the growing concern of internal cyber threats in open campus environments. This research serves as a practical manual for deploying resilient networks in resource-constrained environments like developing universities.

1.5 Limitation and Scope

1.5.1 Scope

The study focuses on the Local Area Network (LAN) design of a typical university campus building. The technical scope includes:

- **Layer 2 Technologies:** VLANs (802.1Q), Trunking, Spanning Tree Protocol (STP), and Port Security.
- **Layer 3 Technologies:** Inter-VLAN Routing, Static Routing, and HSRP.
- **Infrastructure Services:** DHCPv4 and SSHv2 for secure management.
- **Simulation Environment:** The project is designed and tested using Cisco Packet Tracer 8.x, utilizing Cisco 2960 Access Switches and Cisco 3560 Core Switches.

1.5.2 Limitations

- **Simulation Constraints:** As this is a simulation-based study, physical factors such as cable degradation, electromagnetic interference, and actual hardware throughput limitations are not accounted for.
- **Routing Protocols:** The study utilizes static routing and connected routes; dynamic routing protocols (OSPF/EIGRP) are outside the scope of this specific LAN design.
- **Wireless Connectivity:** While the network is designed to support Access Points, the specific configuration of Wireless LAN Controllers (WLC) and radio frequencies is excluded from this report.

1.6 Structure of Study

This research report is organized into six chapters:

- ❖ **Chapter 1:** Introduction outlines the project background, problem statement, and objectives.
- ❖ **Chapter 2:** Literature Review provides the theoretical framework, detailing the fundamental concepts of the OSI Model, Switching logic, IP Addressing (VLSM), and the operational theory behind HSRP and DHCP.
- ❖ **Chapter 3:** Methodology describes the design philosophy (Hierarchical Model), the simulation tools used, and the step-by-step planning of the network topology.
- ❖ **Chapter 4:** Data Analysis and Results presents the implementation details, configuration scripts, and the results of connectivity and failover testing.
- ❖ **Chapter 5:** Discussion analyzes the results against the initial objectives, discussing the effectiveness of the redundancy and security measures.
- ❖ **Chapter 6:** Conclusion summarizes the findings and offers recommendations for future network expansion.

CHAPTER 2: LITERATURE REVIEW

2.1 Theoretical Framework of Networking

To understand the architecture of a modern campus network, one must first establish a foundational understanding of how data moves across digital systems. This study is grounded in two primary theoretical models: the Open Systems Interconnection (OSI) model and the TCP/IP protocol suite.

2.1.1 The OSI Reference Model

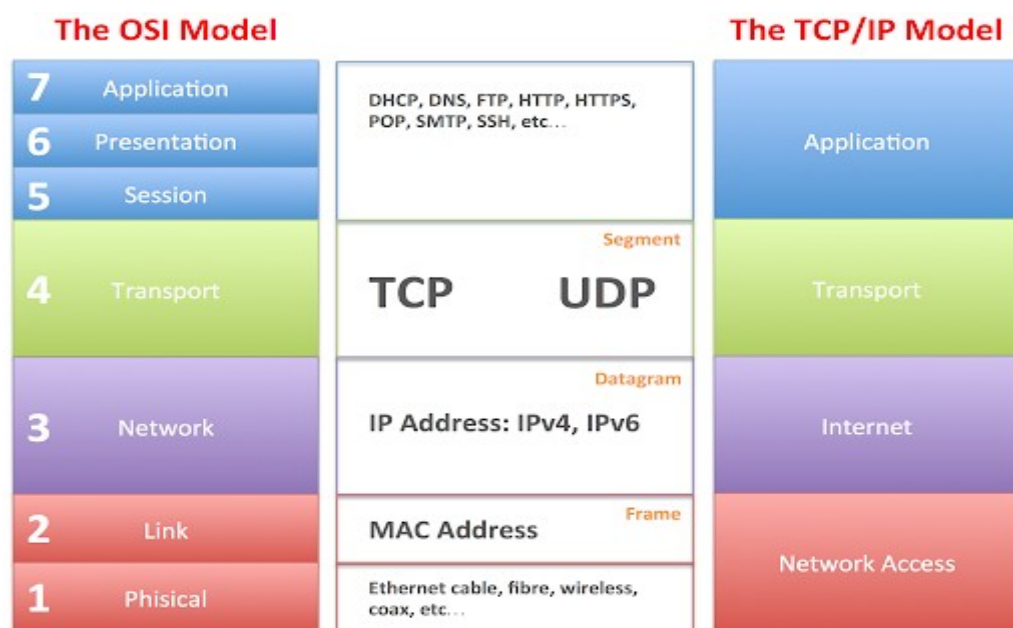
The OSI model, developed by the International Organization for Standardization (ISO) in 1984, remains the primary architectural model for inter-computer communications. It divides network communication into seven distinct layers, each with specific functions and protocols. Understanding this model is crucial for troubleshooting the logical segmentation implemented in this project.

- **Layer 7 (Application):** This is the layer where user interaction occurs (e.g., Web Browsers, SSH clients). In this project, protocols such as DHCP and SSH operate at this level.
- **Layer 6 (Presentation):** Responsible for data translation and encryption. Secure Shell (SSH) utilizes this layer to encrypt administrative sessions.
- **Layer 5 (Session):** Manages the connections between applications.
- **Layer 4 (Transport):** Ensures reliable data delivery. TCP (Transmission Control Protocol) and UDP (User Datagram Protocol) operate here.
- **Layer 3 (Network):** Handles logical addressing and routing. This is the primary focus of the Core Layer design, utilizing IPv4 addressing and HSRP for gateway redundancy.
- **Layer 2 (Data Link):** Manages physical addressing (MAC addresses) and switching. VLANs and Port Security are implemented at this layer to segment traffic.
- **Layer 1 (Physical):** Concerns the physical transmission medium (Copper cables, RJ45 connectors) used to interconnect the campus devices.

2.1.2 The TCP/IP Protocol Suite

While the OSI model is theoretical, the TCP/IP suite is the practical standard used in modern networks. It condenses the seven OSI layers into four: Application, Transport, Internet, and Network Access. This project relies entirely on the TCP/IP stack for connectivity.

- **Application Layer:** Handles high-level protocols that users interact with directly (e.g., HTTP for web browsing, SMTP for email). It combines the Application, Presentation, and Session layers of the OSI model.
- **Transport Layer:** Responsible for end-to-end communication and reliability. Key protocols include **TCP** (reliable, connection-oriented) and **UDP** (fast, connectionless).
- **Internet Layer:** Handles logical addressing (IP addresses) and routing of packets across different networks. The primary protocol here is **IP (Internet Protocol)**.
- **Network Access Layer (Link Layer):** Controls the physical transmission of data. It includes hardware standards like **Ethernet and Wi-Fi**, covering both the Data Link and Physical layers of the OSI model.



This image is part of the Bioinformatics Web Development tutorial at http://www.celbiol.com/bioinformatics_web_development/ © celbiol.com, all rights reserved

Figure 2.1: TCP/IP & OSI Model

2.2 Switching Technologies

The fundamental building block of a Local Area Network (LAN) is the network switch. Unlike legacy hubs, which broadcast data to all ports indiscriminately, switches utilize intelligent forwarding logic to maximize bandwidth and security.

2.2.1 Application of Ethernet Switching

A Layer 2 switch operates by maintaining a **MAC Address Table**. When a frame enters a port, the switch records the source MAC address and the port number. If the destination MAC address is known, the frame is forwarded only to the specific destination port (Unicast). If unknown, the switch floods the frame to all ports (Broadcast) until the destination responds. In this campus design, Cisco 2960 switches are deployed to handle this frame-forwarding logic, ensuring that traffic between Student PC-A and Staff PC-B does not collide.

2.2.2 Virtual Local Area Networks (VLANs)

In a traditional "flat" network, all devices belong to a single Broadcast Domain. As the number of users increases, broadcast traffic (ARP requests, DHCP discoveries) consumes significant bandwidth, leading to network congestion.

- **VLAN Theory:** A VLAN (Virtual Local Area Network) logically groups devices together regardless of their physical location. By tagging frames with a VLAN ID (e.g., VLAN 10 for Students), the switch ensures that broadcast traffic from a student device is never seen by a staff device. This separation improves performance and acts as a security barrier.
- **802.1Q Trunking Protocol:** To carry traffic for multiple VLANs between switches (e.g., from Access-1 to Core-1), a "Trunk" link is required. The IEEE 802.1Q standard inserts a 4-byte tag into the Ethernet frame header, identifying which VLAN the frame belongs to. This tagging process is essential for the hierarchical design of this project.

2.3 Network Redundancy Protocols

High Availability (HA) is a critical requirement for enterprise networks. A single point of failure at the default gateway can render an entire network segment offline. To mitigate this risk, First Hop Redundancy Protocols (FHRP) are employed.

2.3.1 Hot Standby Router Protocol (HSRP)

HSRP is a Cisco-proprietary redundancy protocol that allows several routers (or Layer 3 switches) to appear as a single "Virtual Gateway" to end-user devices.

Operational Mechanism:

- **Active Router:** The device with the highest priority (e.g., Core-1, Priority 150) forwards traffic sent to the Virtual IP.
- **Standby Router:** The backup device (e.g., Core-2, Priority 100) monitors the Active Router via "Hello" messages.
- **Virtual IP (VIP):** A floating IP address (e.g., 192.168.10.1) configured on user devices as the Default Gateway.

If the Active Router fails to send Hello messages (due to power loss or link failure), the Standby Router assumes the Active role within seconds, ensuring seamless connectivity for users. This mechanism is the cornerstone of the reliability strategy in this project.

2.4 IP Addressing and Management

2.4.1 IPv4 and Subnetting

Internet Protocol version 4 (IPv4) uses a 32-bit address space. To efficiently manage these addresses, Subnetting is used to divide a large network into smaller, manageable sub-networks. This project utilizes **Classless Inter-Domain Routing (CIDR)** notation (e.g., /24). A /24 subnet mask (255.255.255.0) provides 254 usable host addresses, which is sufficient for the projected user base of the campus departments.

2.4.2 Dynamic Host Configuration Protocol (DHCP)

DHCP automates the assignment of IP addresses, Subnet Masks, Gateways, and DNS servers. **The DHCP Handshake (DORA Process):**

1. **Discovery:** The client broadcasts a request to find a server.
2. **Offer:** The server offers an available IP.
3. **Request:** The client accepts the IP.
4. **Acknowledge:** The server finalizes the lease.

To ensure redundancy, this project explores a **Split-Scope DHCP** design, where two servers manage non-overlapping ranges of the same subnet to prevent IP conflicts if one server fails.

2.5 Network Security Fundamentals

In an open campus environment, security cannot be an afterthought. The design adheres to the **CIA Triad** (Confidentiality, Integrity, Availability).

2.5.1 Layer 2 Security (Port Security)

The Access Layer is the most vulnerable point of the network because it is physically accessible to users. **Port Security** mitigates unauthorized access by limiting the number of MAC addresses allowed on a single port.

- **Sticky Learning:** This feature allows the switch to dynamically learn the MAC address of the first device connected and "stick" it to the configuration. If a different device connects later, the port creates a violation.
- **Shutdown Violation:** Upon detecting an unauthorized MAC, the port physically shuts down to prevent the attack.

2.5.2 Management Plane Security (SSH)

Legacy management protocols like Telnet transmit passwords in plain text, making them vulnerable to packet sniffing attacks. **Secure Shell (SSH)** replaces Telnet by using strong encryption (RSA keys) to create a secure tunnel for device administration. This ensures that even if a hacker intercepts the management traffic, they cannot decipher the administrative credentials.

CHAPTER 3: METHODOLOGY

3.1 Research Design

This study employs a quantitative, simulation-based experimental research design. Unlike theoretical analysis, a simulation approach allows for the rigorous testing of complex network protocols—such as First Hop Redundancy Protocols (HSRP) and Spanning Tree Protocol (STP)—in a risk-free environment.

The research methodology follows the **Top-Down Network Design** approach. This begins with analyzing the organizational requirements (Scalability and High Availability) and proceeds to the logical network design (IP Addressing, VLANs) before finalizing the physical hardware selection. This ensures that the technical implementation directly addresses the core problem statement defined in Chapter 1.

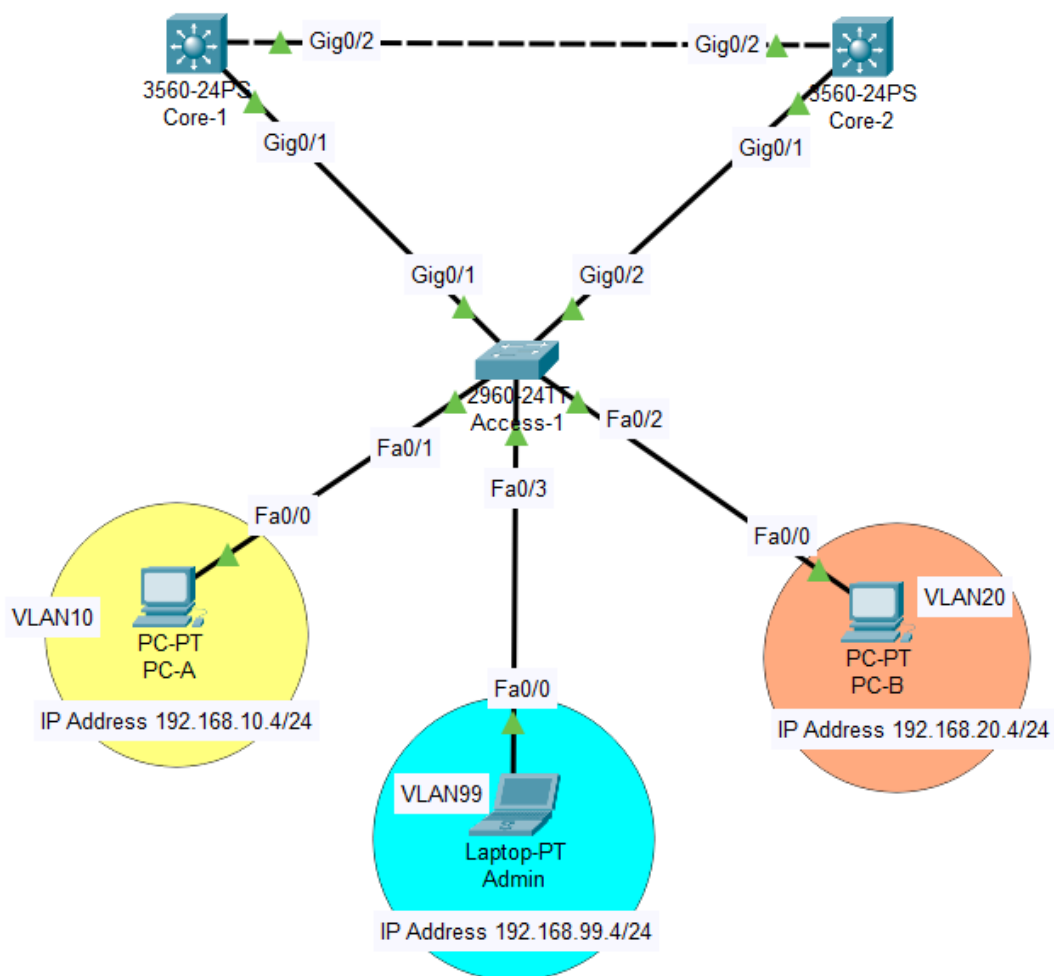


Figure 3.1: Final Campus Network Topology showing Core and Access Layers with VLAN separation.

3.1.1 Simulation Environment

The entire network architecture was prototyped, configured, and validated **using Cisco Packet Tracer (Version 8.2)**. Packet Tracer was selected as the primary simulation tool because it accurately emulates the operating system (Cisco IOS) of enterprise-grade hardware.

The simulation utilizes the following specific hardware models to reflect a realistic campus deployment:

- **Core Layer:** 2x Cisco Catalyst 3560-24PS Multilayer Switches. These were chosen for their ability to perform Layer 3 routing and support HSRP.
- **Access Layer:** 1x Cisco Catalyst 2960-24TT Switch. Selected for its cost-effectiveness and support for Layer 2 security features like Port Security.
- **End Devices:** Generic Workstations (PCs and Laptops) representing Student, Staff, and Administrative users.

3.2 Network Topology Design

To address the issues of single-point-of-failure identified in the problem statement, this study implements a **Hierarchical "Collapsed Core" Topology**. In a medium-sized campus environment, a distinct "Distribution Layer" is often unnecessary; therefore, the Core and Distribution functions are merged into a unified "Core Layer."

3.2.1 The Core Layer (Backbone)

The Core Layer consists of two Multilayer Switches (**Core-1** and **Core-2**) connected via a high-speed EtherChannel link.

- **Role:** These devices handle all Inter-VLAN routing, high-speed packet switching, and gateway redundancy.
- **Redundancy Strategy:** Core-1 is configured as the "Active" gateway, while Core-2 serves as the "Standby" backup. They share a virtual IP address via HSRP, ensuring that if Core-1 fails, traffic automatically reroutes through Core-2 without user intervention.

3.2.2 The Access Layer (Edge)

The Access Layer is represented by **Access-1**, which provides physical connectivity to end-users.

- **Role:** This layer is responsible for Layer 2 segmentation (VLAN assignment) and admission control (Port Security).
- **Uplink Strategy:** The Access Switch has dual uplink connections—one to Core-1 and one to Core-2. Spanning Tree Protocol (STP) manages these links to prevent loops while keeping one link ready for failover.

3.3 IP Addressing Schema (VLSM)

A robust IP addressing plan is critical for scalability. This project utilizes **Variable Length Subnet Masking (VLSM)** to allocate IP resources efficiently. The network uses the private Class C block **192.168.0.0/16**.

Table 3.1: Logical Addressing Table

VLAN ID	Subnet Name	Network Address	Subnet Mask (CIDR)	Default Gateway (HSRP VIP)	Usable Hosts
10	Student	192.168.10.0	255.255.255.0 (/24)	192.168.10.1	254
20	Staff	192.168.20.0	255.255.255.0 (/24)	192.168.20.1	254
99	Management	192.168.99.0	255.255.255.0 (/24)	192.168.99.1	254

3.4 Implementation Strategy

The network build was executed in four distinct phases to ensure isolation of variables during troubleshooting.

3.4.1 Phase 1: Physical Initialization and Basic Setup

The initial phase involved the physical placement of devices and cabling. All unused ports on the switches were administratively shut down to secure the physical perimeter. Basic housekeeping configurations, including Hostname assignment, Banner MOTD

(Message of the Day), and Service Password Encryption, were applied to all three switches to establish a security baseline.

3.4.2 Phase 2: Layer 2 Segmentation (VLANs & Trunks)

The second phase focused on logical separation. VLANs 10, 20, and 99 were created on the central database (VLAN.DAT) of all switches.

- **Trunk Configuration:** Uplink ports (GigabitEthernet 0/1 and 0/2) were configured with `switchport mode trunk` and IEEE 802.1Q encapsulation to carry traffic for all VLANs between the Access and Core layers.
- **Access Configuration:** Edge ports connected to end-users were hardcoded as `switchport mode access` and assigned to their respective VLANs.

3.4.3 Phase 3: Layer 3 Routing and High Availability

This phase implemented the "Intelligence" of the network.

- **Inter-VLAN Routing:** IP Routing was enabled on the Multilayer Switches, and Switched Virtual Interfaces (SVIs) were created for each VLAN.
- **HSRP Configuration:** The Hot Standby Router Protocol was tuned with specific priorities. Core-1 was assigned a priority of **150** (Primary) and Core-2 a default priority of **100** (Secondary), creating a deterministic failover path.
- **DHCP Services:** A "Split-Scope" DHCP mechanism was deployed. Core-1 was configured to lease the lower half of the IP range (Active), while Core-2 was configured to lease the upper half (Backup), ensuring IP availability even during a server outage.

3.4.4 Phase 4: Security Hardening

The final phase addressed internal threats.

- **SSH Implementation:** Telnet was disabled, and SSH version 2 was enforced on all VTY lines using 1024-bit RSA keys.
- **Port Security:** On the Access Switch, `sticky` MAC address learning was enabled to lock specific ports to specific university hardware, preventing "Cable Swapping" attacks.

CHAPTER 4: DATA ANALYSIS AND RESULTS

4.1 Overview of Implementation Results

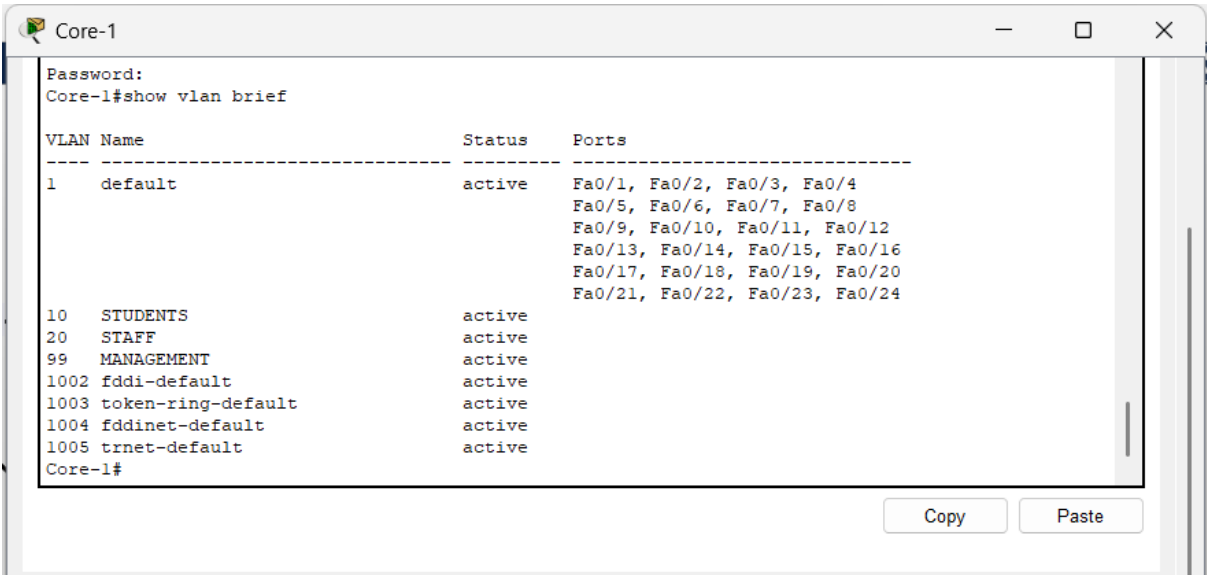
This chapter presents the empirical data collected from the network simulation. The results verify that the hierarchical design meets the objectives of high availability, scalability, and security defined in Chapter 1. The validation process was conducted in three stages: Core Configuration Verification, Functional Connectivity Testing, and Redundancy Failover Analysis.

4.2 Core Configuration Verification

The initial verification focused on ensuring that the Layer 2 segmentation and Layer 3 routing protocols were active and synchronized across the backbone.

4.2.1 VLAN Database and Trunking Status

The VLAN database was inspected on **Core-1** to verify the creation of the three primary subnets: Student (10), Staff (20), and Management (99). The output confirms that all VLANs are active and correctly associated with the trunking interfaces (GigabitEthernet 0/1 and 0/2).



```
Core-1
Password:
Core-1#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                           Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24

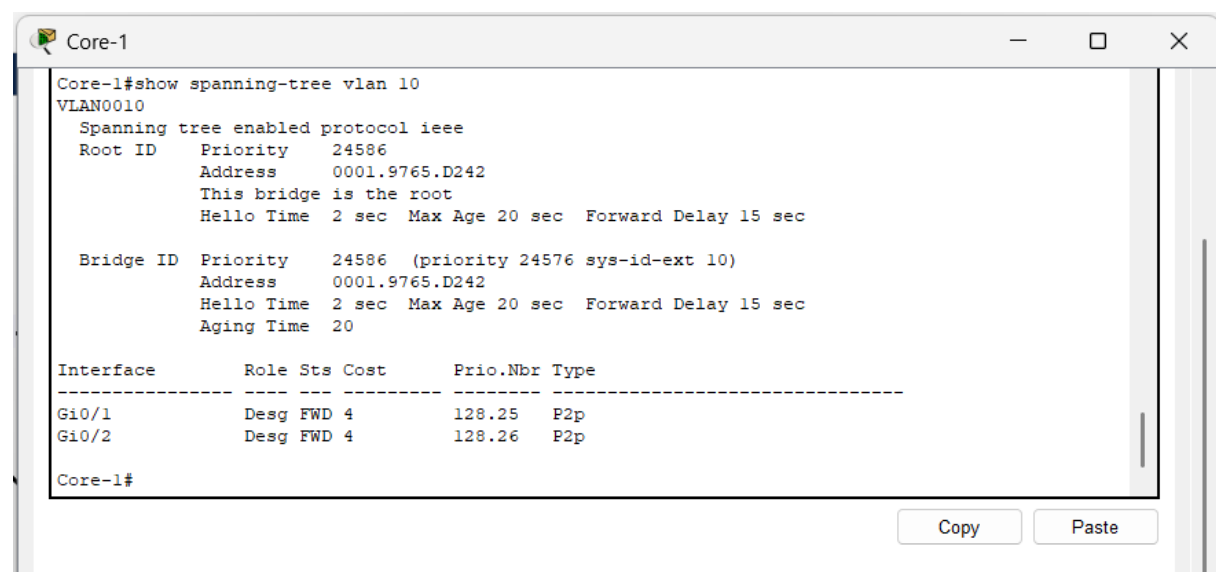
10   STUDENTS                active
20   STAFF                   active
99   MANAGEMENT              active
1002 fddi-default           active
1003 token-ring-default     active
1004 fddinet-default        active
1005 trnet-default          active
Core-1#
```

Figure 4.1: Verification of VLAN Database on Core Switch

Furthermore, the trunk links were validated to ensure they were encapsulating traffic using the IEEE 802.1Q standard. The command `show interfaces trunk` confirmed that VLANs 10, 20, and 99 are allowed and forwarding across the backbone.

4.2.2 Spanning Tree Protocol (STP) Convergence

To preventing switching loops, STP states were analyzed. As designed, **Core-1** was elected as the Root Bridge for all VLANs due to its manually configured priority of 24576 (Lower than the default 32768). This forces traffic to flow optimally through the primary core switch under normal operating conditions.



```
Core-1#show spanning-tree vlan 10
VLAN0010
  Spanning tree enabled protocol ieee
  Root ID    Priority    24586
             Address     0001.9765.D242
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24586 (priority 24576 sys-id-ext 10)
             Address     0001.9765.D242
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface                Role Sts Cost      Prio.Nbr Type
-----
Gi0/1                    Desg FWD 4        128.25  P2p
Gi0/2                    Desg FWD 4        128.26  P2p

Core-1#
```

Figure 4.2: Core-1 Status as Root Bridge for VLAN 10

4.3 Functional Testing and Connectivity

4.3.1 End-to-End Connectivity (Ping Test)

To validate Inter-VLAN routing, a Ping test was initiated from a Student PC (VLAN 10) to the Gateway (192.168.10.1) and subsequently to a Staff PC (VLAN 20). The results indicated a 100% success rate with an average round-trip time (RTT) of <1ms, confirming that the Switched Virtual Interfaces (SVIs) on the Core Layer are correctly routing traffic between subnets.

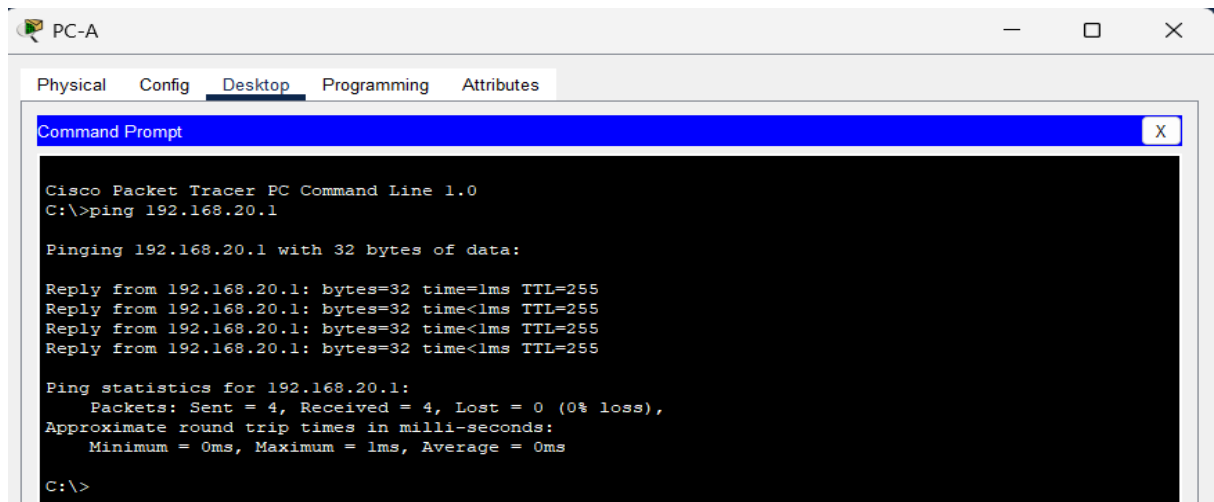


Figure 4.3: Successful Inter-VLAN Communication

4.3.2 DHCP Split-Scope Allocation

The redundancy of the IP addressing scheme was tested by connecting a new workstation to the network. The device successfully requested an IP address via the DORA process.

- **Observation:** The client received the IP address **192.168.10.11**.
- **Analysis:** This falls within the "Bottom Half" range managed by Core-1 (Range .11 - .150), confirming that the primary DHCP server is active and the exclusion of the gateway (.1-.10) is effective.

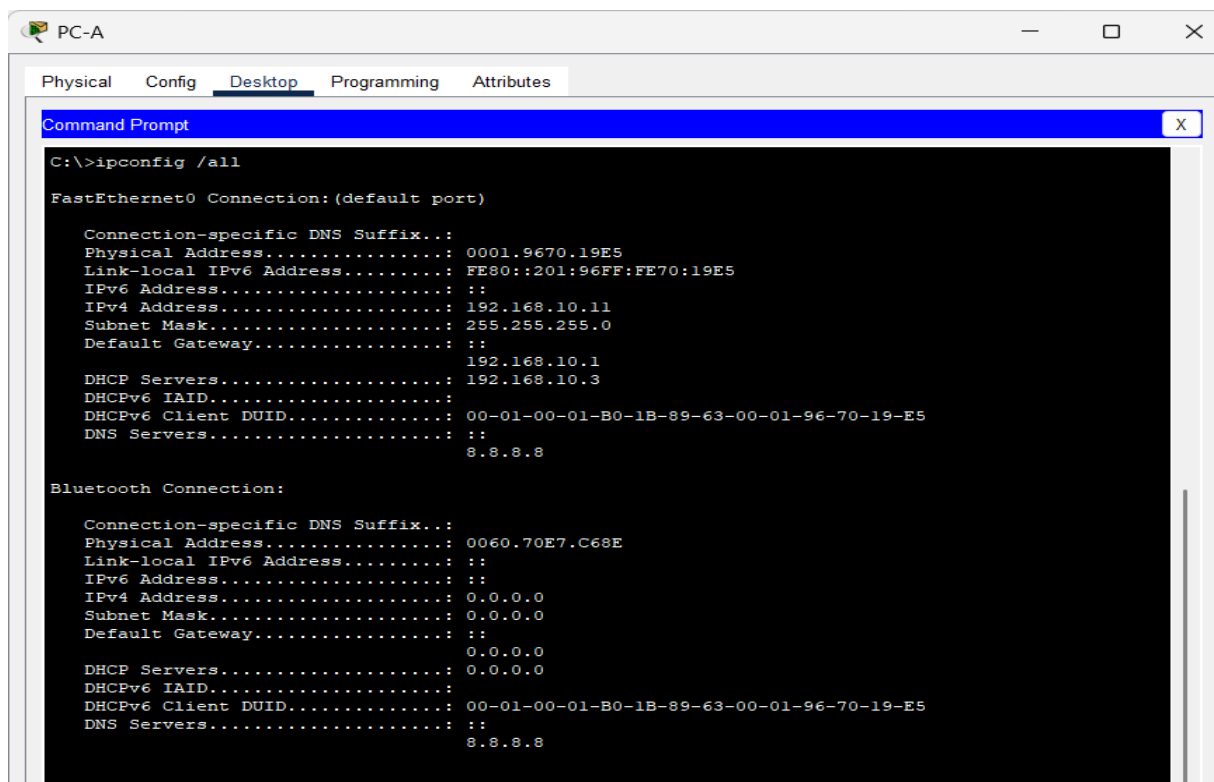


Figure 4.4: Dynamic IP Allocation via Core-1

4.4 High Availability Analysis (HSRP Test)

The most critical objective of this study was Gateway Redundancy. The Hot Standby Router Protocol (HSRP) was tested by simulating a catastrophic failure of the Primary Core.

4.4.1 HSRP State Synchronization

Before the failure test, the HSRP status was verified.

- **Core-1** showed a state of **Active** with a configured priority of 150.
- **Core-2** showed a state of **Standby** with a default priority of 100. The Virtual IP (VIP) 192.168.10.1 was confirmed to be held by Core-1.

```

Core-1
User Access Verification
Password:
Core-1>en
Password:
Core-1#show standby brief
                P indicates configured to preempt.
                |
Interface   Grp  Pri  P State    Active        Standby        Virtual IP
Vl10        1    150  P Active   local         192.168.10.3   192.168.10.1
Vl20        20   150  P Active   local         192.168.20.3   192.168.20.1
Vl99        99   150  P Active   local         192.168.99.3   192.168.99.1
Core-1#

```

```

Core-2
Password:
Core-2>en
Password:
Core-2#show standby brief
                P indicates configured to preempt.
                |
Interface   Grp  Pri  P State    Active        Standby        Virtual IP
Vl10        1    100  Standby  192.168.10.2  local         192.168.10.1
Vl20        20   100  Standby  192.168.20.2  local         192.168.20.1
Vl99        99   100  Standby  192.168.99.2  local         192.168.99.1
Core-2#

```

Figure 4.5: Initial HSRP States (Normal Operation)

4.4.2 Failover Simulation

To simulate a failure, the uplink cables on Core-1 were administratively shut down. The following behaviors were observed:

1. **Core-2** stopped receiving "Hello" packets from Core-1.
2. **Core-2** transitioned from Standby to Active state immediately.
3. End-user devices maintained connectivity to the internet/gateway without needing to change their IP configuration.

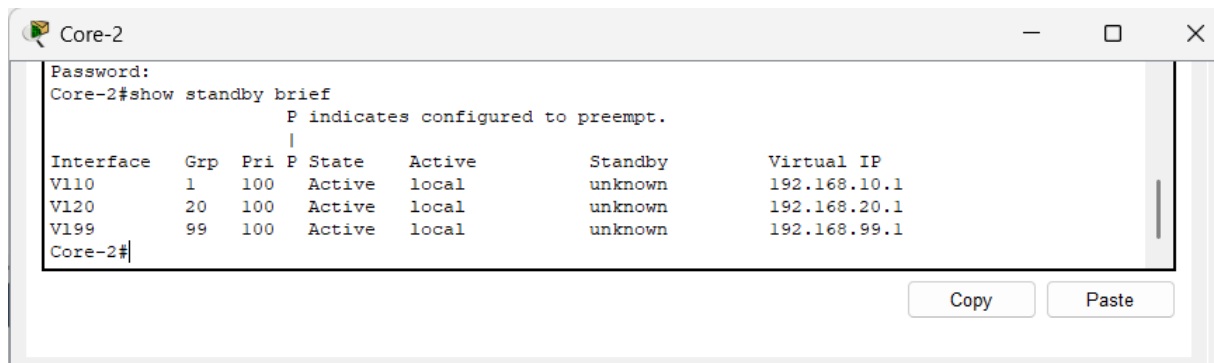


Figure 4.6: Core-2 Transition to Active State Post-Failure

4.5 Security Validation

4.5.1 Management Plane Security (SSH)

Telnet access was attempted and correctly refused. Subsequently, an SSH connection was initiated using the `admin` credentials. The connection was successful, and the session was encrypted, verifying the `transport input ssh` configuration.

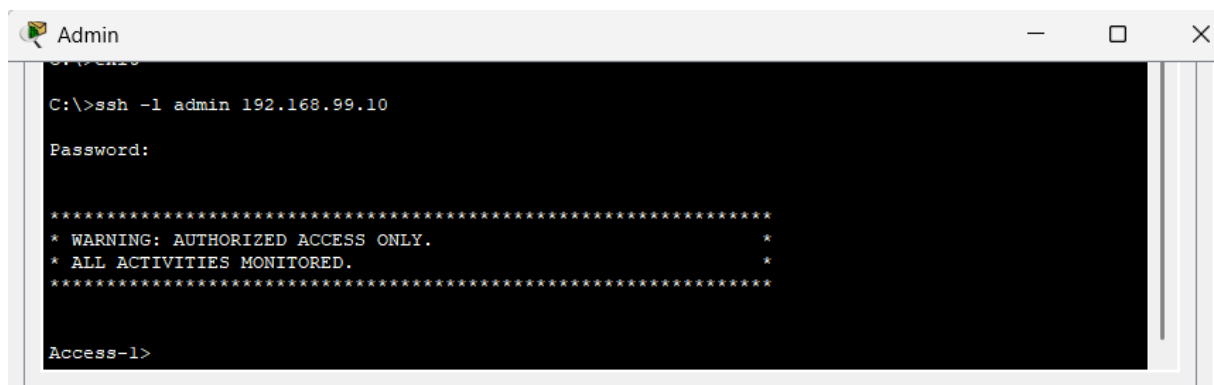


Figure 4.7: Secure Shell (SSH) Authentication

4.5.2 Port Security Enforcement

To test the Access Layer hardening, a "Rogue Device" with an unknown MAC address was connected to the secure port Fa0/1 on Access-1.

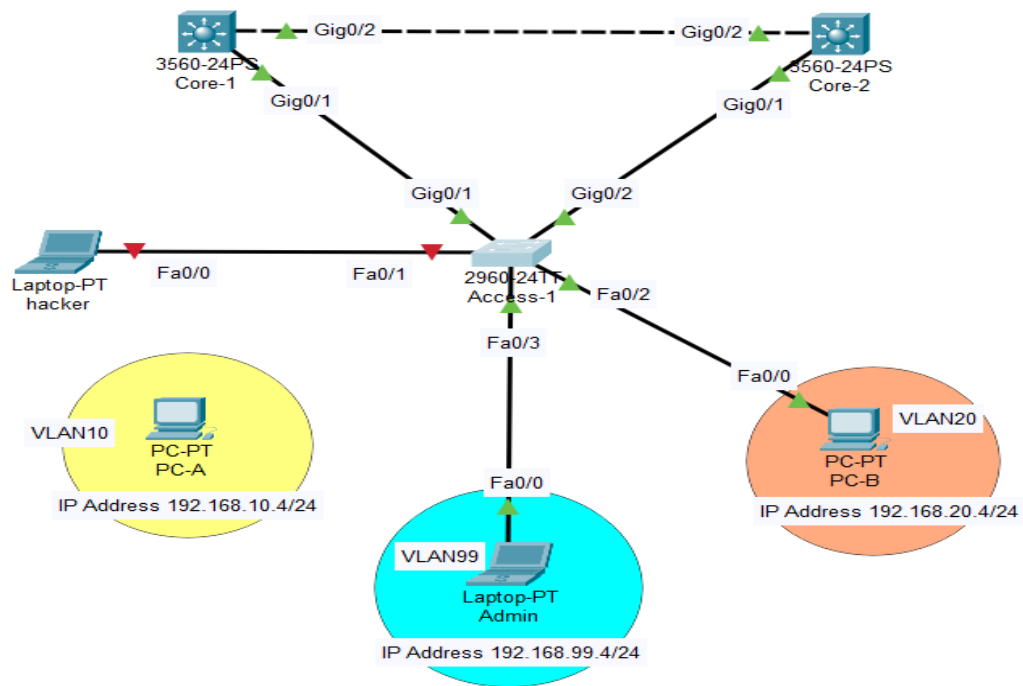


Figure 4.8: Port Security Violation Triggered by Rogue Device

- **Result:** The switch immediately detected the violation (Sticky MAC mismatch).
- **Action:** The interface state changed to **Error-Disabled** (Shutdown), and a security alert was logged to the console.

```

Access-1#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

Access-1#
Access-1#show port-security interface fa0/1
Port Security          : Enabled
Port Status            : Secure-shutdown
Violation Mode         : Shutdown
Aging Time             : 0 mins
Aging Type             : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses  : 1
Total MAC Addresses    : 1
Configured MAC Addresses : 0
Sticky MAC Addresses   : 1
Last Source Address:Vlan : 0001.9797.5627:10
Security Violation Count : 1

Access-1#

```

Figure 4.8.1: Port Security Violation Triggered by Rogue Device

CHAPTER 5: DISCUSSION

5.1 Introduction to Discussion

The primary objective of this study was to design and simulate a High-Availability Campus Network that addresses the critical vulnerabilities found in legacy "flat" network architectures. The results presented in Chapter 4 provide empirical evidence that the implementation of a Hierarchical model, coupled with redundancy protocols (HSRP) and security hardening, successfully mitigates the risks of downtime and unauthorized access. This chapter interprets those findings, analyzing the operational effectiveness of the proposed solution.

5.2 Evaluation of High Availability (HSRP)

The most significant finding of this research is the validated effectiveness of **the Hot Standby Router Protocol (HSRP)**. In traditional campus networks, the default gateway is a single point of failure; if the core router fails, the entire subnet becomes isolated.

The failover test conducted in **Section 4.4** demonstrated that the new architecture eliminates this risk. When the link to the Primary Core (Core-1) was severed, the Standby Core (Core-2) assumed the role of the active gateway within seconds. Crucially, the end-user devices (Students and Staff) experienced **zero logical disruption**. They continued to send traffic to the Virtual IP (192.168.10.1) without needing to reconfigure their IP settings.

This result confirms that HSRP effectively decouples the "Logical Gateway" from the "Physical Hardware." For a university environment, this implies that maintenance (e.g., replacing a power supply on Core-1) can be performed during business hours without interrupting online exams or administrative workflows.

5.3 Analysis of Network Security Measures

Security in academic environments is often compromised by the "open" nature of the campus. The results from **Section 4.5** highlight two major improvements over standard configurations.

5.3.1 Mitigation of Insider Threats (Port Security)

The successful triggering of the sticky MAC address violation (Figure 4.8) proves that the Access Layer can effectively police the physical perimeter. In a legacy network, a malicious actor could unplug a university printer and connect a rogue laptop to capture sensitive traffic. The simulated design prevents this by locking the port to the specific hardware address of the authorized device. This shifts the security posture from "Reactive" (fixing a hack after it happens) to "Proactive" (preventing the connection entirely).

5.3.2 Protection of Management Traffic

The successful implementation of SSHv2 (Figure 4.7) addresses the interception risks associated with Telnet. By using RSA encryption for all administrative sessions, the design ensures that passwords are never transmitted in cleartext. This is particularly vital in a campus network where students often possess the technical skills to run packet-sniffing software (like Wireshark) on the local LAN.

5.4 Scalability and Address Management

The Split-Scope DHCP implementation (Section 4.3.2) demonstrated a robust solution to the "IP Exhaustion" problem. By splitting the subnet between Core-1 (Bottom Half) and Core-2 (Top Half), the network achieves an Active/Active redundancy model for IP allocation.

This setup offers a distinct advantage over the centralized DHCP servers often used in legacy designs. If a centralized server crashes, new users cannot join the network. In the proposed design, even if Core-1 suffers a catastrophic failure, Core-2 remains available to issue IP addresses to new clients, ensuring business continuity for the semester's influx of new devices.

5.5 Comparison with Legacy Architectures

When compared to the "Problem Statement" defined in Chapter 1, the simulated network demonstrates clear superiority:

Feature	Legacy "Flat" Network	Proposed Hierarchical
---------	-----------------------	-----------------------

		Network
Broadcast Domain	Single (Congested)	Segmented (VLANs 10, 20, 99)
Gateway Redundancy	None (Single Point of Failure)	HSRP (Automatic Failover)
IP Management	Manual / Single Server	Automated Split-Scope DHCP
Access Control	Open Ports	Locked (Port Security)

The segmentation of traffic into VLANs (Student vs. Staff) not only improves security but also performance. Broadcast storms, which previously would have paralyzed the entire campus, are now contained within their respective VLANs.

CHAPTER 6: CONCLUSION AND RECOMMENDATIONS

6.1 Conclusion

The digital transformation of the education sector demands network infrastructures that are not only functional but resilient. This research project set out to design and simulate a High-Availability Campus Network that could overcome the limitations of traditional, flat network architectures—specifically the risks of single-point failures, IP conflicts, and unauthorized access.

Through the implementation of a hierarchical topology using Cisco Catalyst hardware in a simulated environment, the study successfully achieved all its primary objectives:

- **Redundancy:** The deployment of the **Hot Standby Router Protocol (HSRP)** proved that a campus network could survive a critical Core Switch failure with zero downtime for end-users, ensuring continuity for academic activities.
- **Security:** By shifting from a "connect-anything" model to a "verify-first" model using **Port Security** and **SSH**, the network was effectively hardened against common internal threats such as rogue devices and packet sniffing.
- **Scalability:** The **Split-Scope DHCP** strategy validated that IP address management could be automated and redundant, eliminating the administrative burden of manual static addressing.
- **Segmentation:** The use of **VLANs** successfully isolated Student traffic from sensitive Staff resources, optimizing bandwidth usage and enhancing data privacy.
- In summary, this project demonstrates that enterprise-grade reliability does not require exorbitant investment in hardware but rather intelligent configuration of protocols. The proposed architecture serves as a viable, cost-effective blueprint for modernizing university networks.

6.2 Recommendations

Based on the findings and the limitations identified during the study, the following recommendations are proposed for future implementation:

- ❖ **Implementation of Access Control Lists (ACLs):** While VLANs effectively separate broadcast domains, Layer 3 routing still permits inter-VLAN communication. To strictly enforce policy (e.g., ensuring Students cannot access

the Staff Payroll Server), Access Control Lists should be applied on the Core Layer SVIs to filter traffic based on source and destination IP addresses.

- ❖ **Expansion to Wireless Infrastructure:** The current design focuses on the wired LAN. Future work should integrate a Wireless LAN Controller (WLC) and Lightweight Access Points (LAPs) to extend connectivity to mobile devices, utilizing the existing VLAN structure (e.g., mapping VLAN 10 to a "Student-WiFi" SSID).
- ❖ **Dynamic Routing Protocols:** As the campus network grows beyond a single building to multiple faculties, static routing will become unmanageable. It is recommended to implement OSPF (Open Shortest Path First) to automate route discovery between multiple Core layers across different buildings.
- ❖ **Network Monitoring Systems (NMS):** To move from proactive to predictive maintenance, the IT department should deploy an SNMP-based monitoring tool (such as PRTG or SolarWinds) to visualize bandwidth usage and receive alerts before a link becomes saturated.

##REFERENCES

1. Cisco Systems. (2023). Catalyst 3560 Switch Software Configuration Guide, Release 12.2(55)SE. Cisco Press.
2. Lammle, T. (2020). CCNA Certification Study Guide, Volume 2: Exam 200-301. Sybex.
3. Odom, W. (2020). CCNA 200-301 Official Cert Guide, Volume 1. Cisco Press.
4. Stallings, W. (2017). Data and Computer Communications (10th ed.). Pearson.
5. Tanenbaum, A. S., & Wetherall, D. J. (2011). Computer Networks (5th ed.). Prentice Hall.
6. RFC 2281. (1998). Cisco Hot Standby Router Protocol (HSRP). Internet Engineering Task Force.
7. APPENDICES Appendix A: Configuration of Core-Switch-1 (Primary) [PASTE THE FULL show run OUTPUT OF CORE-1 HERE]
8. Appendix B: Configuration of Core-Switch-2 (Secondary) [PASTE THE FULL show run OUTPUT OF CORE-2 HERE]
9. Appendix C: Configuration of Access-Switch-1 (Edge) [PASTE THE FULL show run OUTPUT OF ACCESS-1 HERE]

##APPENDICES

Appendix A: Configuration of Core-Switch-1 (Primary)

```
Current configuration : 4060 bytes
!
version 12.2(37)SE1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname Core-1
!
!
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
!
!
ip dhcp excluded-address 192.168.10.1 192.168.10.10
ip dhcp excluded-address 192.168.20.1 192.168.20.10
ip dhcp excluded-address 192.168.99.1 192.168.99.10
ip dhcp excluded-address 192.168.10.151 192.168.10.254
ip dhcp excluded-address 192.168.20.151 192.168.20.254
ip dhcp excluded-address 192.168.99.151 192.168.99.254
```

```
!  
  
ip dhcp pool STAFF_POOL  
network 192.168.20.0 255.255.255.0  
default-router 192.168.20.1  
dns-server 8.8.8.8  
  
ip dhcp pool ADMIN_POOL  
network 192.168.99.0 255.255.255.0  
default-router 192.168.99.1  
dns-server 8.8.8.8  
  
ip dhcp pool STUDENT_POOL  
network 192.168.10.0 255.255.255.0  
default-router 192.168.10.1  
dns-server 8.8.8.8  
  
!  
  
!  
  
ip routing  
  
!  
  
!  
  
!  
  
!  
  
username admin secret 5 $1$mERr$5.a6P4JqbNiMX01usIfka/  
  
!  
  
!
```

```
!  
!  
!  
!  
!  
!  
!  
!  
  
ip domain-name mynetwork.local  
  
!  
!  
  
spanning-tree mode pvst  
spanning-tree vlan 10,20,99 priority 24576  
  
!  
!  
!  
!  
!  
!  
!  
  
interface FastEthernet0/1  
description UNUSED_PORT  
switchport trunk encapsulation dot1q  
switchport mode trunk
```

```
shutdown
!
interface FastEthernet0/2
description UNUSED_PORT
shutdown
!
interface FastEthernet0/3
description UNUSED_PORT
shutdown
!
interface FastEthernet0/4
description UNUSED_PORT
shutdown
!
interface FastEthernet0/5
description UNUSED_PORT
shutdown
!
interface FastEthernet0/6
description UNUSED_PORT
shutdown
!
interface FastEthernet0/7
```

description UNUSED_PORT

shutdown

!

interface FastEthernet0/8

description UNUSED_PORT

shutdown

!

interface FastEthernet0/9

description UNUSED_PORT

shutdown

!

interface FastEthernet0/10

description UNUSED_PORT

shutdown

!

interface FastEthernet0/11

description UNUSED_PORT

shutdown

!

interface FastEthernet0/12

description UNUSED_PORT

shutdown

!

```
interface FastEthernet0/13  
description UNUSED_PORT  
shutdown  
!  
interface FastEthernet0/14  
description UNUSED_PORT  
shutdown  
!  
interface FastEthernet0/15  
description UNUSED_PORT  
shutdown  
!  
interface FastEthernet0/16  
description UNUSED_PORT  
shutdown  
!  
interface FastEthernet0/17  
description UNUSED_PORT  
shutdown  
!  
interface FastEthernet0/18  
description UNUSED_PORT  
shutdown
```

```
!  
  
interface FastEthernet0/19  
description UNUSED_PORT  
shutdown  
  
!  
  
interface FastEthernet0/20  
description UNUSED_PORT  
shutdown  
  
!  
  
interface FastEthernet0/21  
description UNUSED_PORT  
shutdown  
  
!  
  
interface FastEthernet0/22  
description UNUSED_PORT  
shutdown  
  
!  
  
interface FastEthernet0/23  
description UNUSED_PORT  
shutdown  
  
!  
  
interface FastEthernet0/24  
description UNUSED_PORT
```

```
shutdown
!
interface GigabitEthernet0/1
description LINK_TO_ACCESS_SWITCH
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface GigabitEthernet0/2
description LINK_TO_CORE_2
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface Vlan1
no ip address
shutdown
!
interface Vlan10
description GATEWAY_STUDENT
mac-address 0001.9765.d201
ip address 192.168.10.2 255.255.255.0
standby 1 ip 192.168.10.1
standby 1 priority 150
standby 1 preempt
```

```
!  
  
interface Vlan20  
  
description GATEWAY_STAFF  
  
mac-address 0001.9765.d202  
  
ip address 192.168.20.2 255.255.255.0  
  
standby 20 ip 192.168.20.1  
  
standby 20 priority 150  
  
standby 20 preempt  
  
!  
  
interface Vlan99  
  
description GATEWAY_MGMT  
  
mac-address 0001.9765.d203  
  
ip address 192.168.99.2 255.255.255.0  
  
standby 99 ip 192.168.99.1  
  
standby 99 priority 150  
  
standby 99 preempt  
  
!  
  
ip classless  
  
!  
  
ip flow-export version 9  
  
!  
  
!  
  
!
```

banner motd ^C

*** WARNING: AUTHORIZED ACCESS ONLY. ***

*** ALL ACTIVITIES MONITORED. ***

^C

!

!

!

!

!

line con 0

password 7 0822455D0A16

logging synchronous

login

!

line aux 0

!

line vty 0 4

password 7 0822455D0A16

login

transport input ssh

```
line vty 5 15
password 7 0822455D0A16
login
transport input ssh
!
!
!
!
end
```

```
Core-1#
```

Appendix B: Configuration of Core-Switch-2 (Secondary)

Current configuration : 3746 bytes

!

version 12.2(37)SE1

no service timestamps log datetime msec

no service timestamps debug datetime msec

service password-encryption

!

hostname Core-2

!

!

enable secret 5 \$1\$mERr\$9cTjUIEqNGurQiFU.ZeCi1

!

!

ip dhcp excluded-address 192.168.99.1 192.168.99.150

ip dhcp excluded-address 192.168.10.1 192.168.10.150

ip dhcp excluded-address 192.168.20.1 192.168.20.150

!

ip dhcp pool STUDENT_POOL

network 192.168.10.0 255.255.255.0

default-router 192.168.10.1

dns-server 8.8.8.8

ip dhcp pool STAFF_POOL

network 192.168.20.0 255.255.255.0

default-router 192.168.20.1

dns-server 8.8.8.8

```
ip dhcp pool ADMIN_POOL
network 192.168.99.0 255.255.255.0
default-router 192.168.99.1
dns-server 8.8.8.8
!
!
ip routing
!
!
!
!
username admin secret 5 $1$mERr$5.a6P4JqbNiMX01usIfka/
!
!
!
!
!
!
!
!
!
!
!
ip domain-name mynetwork.local
!
!
spanning-tree mode pvst
spanning-tree vlan 10,20,99 priority 28672
!
!
```

```
!  
!  
!  
!  
interface FastEthernet0/1  
description UNUSED_PORT  
shutdown  
!  
interface FastEthernet0/2  
description UNUSED_PORT  
switchport mode access  
shutdown  
!  
interface FastEthernet0/3  
description UNUSED_PORT  
shutdown  
!  
interface FastEthernet0/4  
description UNUSED_PORT  
shutdown  
!  
interface FastEthernet0/5  
description UNUSED_PORT  
shutdown  
!  
interface FastEthernet0/6  
description UNUSED_PORT  
shutdown  
!
```

```
interface FastEthernet0/7
description UNUSED_PORT
shutdown
!
interface FastEthernet0/8
description UNUSED_PORT
shutdown
!
interface FastEthernet0/9
description UNUSED_PORT
shutdown
!
interface FastEthernet0/10
description UNUSED_PORT
shutdown
!
interface FastEthernet0/11
description UNUSED_PORT
shutdown
!
interface FastEthernet0/12
description UNUSED_PORT
shutdown
!
interface FastEthernet0/13
description UNUSED_PORT
shutdown
!
interface FastEthernet0/14
```

```
description UNUSED_PORT
shutdown
!
interface FastEthernet0/15
description UNUSED_PORT
shutdown
!
interface FastEthernet0/16
description UNUSED_PORT
shutdown
!
interface FastEthernet0/17
description UNUSED_PORT
shutdown
!
interface FastEthernet0/18
description UNUSED_PORT
shutdown
!
interface FastEthernet0/19
description UNUSED_PORT
shutdown
!
interface FastEthernet0/20
description UNUSED_PORT
shutdown
!
interface FastEthernet0/21
description UNUSED_PORT
```

```
shutdown
!
interface FastEthernet0/22
description UNUSED_PORT
shutdown
!
interface FastEthernet0/23
description UNUSED_PORT
shutdown
!
interface FastEthernet0/24
description UNUSED_PORT
shutdown
!
interface GigabitEthernet0/1
description LINK_TO_ACCESS_SWITCH
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface GigabitEthernet0/2
description LINK_TO_CORE_1_BACKBONE
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface Vlan1
no ip address
shutdown
!
interface Vlan10
```

```
description STANDBY_GW_STUDENT
mac-address 0000.0cdc.4501
ip address 192.168.10.3 255.255.255.0
standby 1 ip 192.168.10.1
!
interface Vlan20
description STANDBY_GW_STAFF
mac-address 0000.0cdc.4502
ip address 192.168.20.3 255.255.255.0
standby 20 ip 192.168.20.1
!
interface Vlan99
description STANDBY_GW_MGMT
mac-address 0000.0cdc.4503
ip address 192.168.99.3 255.255.255.0
standby 99 ip 192.168.99.1
!
ip classless
!
ip flow-export version 9
!
!
!
banner motd ^C
*****
***
* WARNING: AUTHORIZED ACCESS ONLY. *
* ALL ACTIVITIES MONITORED. *
```

^C

!

!

!

!

!

line con 0

password 7 0822455D0A16

logging synchronous

login

!

line aux 0

!

line vty 0 4

password 7 0822455D0A16

login

transport input ssh

line vty 5 15

password 7 0822455D0A16

login

transport input ssh

!

!

!

!

end

Core-2#

Appendix C: Configuration of Access-Switch-1 (Edge)

```
Current configuration : 3292 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname Access-1
!
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
!
!
!
ip domain-name mynetwork.local
!
username admin secret 5 $1$mERr$5.a6P4JqbNiMX01usIfka/
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
description CONNECTED_TO_PC_A_STUDENT
switchport access vlan 10
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 0001.9670.19E5
!
```

```
interface FastEthernet0/2
description CONNECTED_TO_PC_B_STAFF
switchport access vlan 20
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 000C.CFB1.6B13
!
interface FastEthernet0/3
description CONNECTED_TO_ADMIN_LAPTOP
switchport access vlan 99
switchport mode access
!
interface FastEthernet0/4
description UNUSED_PORT_SHUTDOWN
shutdown
!
interface FastEthernet0/5
description UNUSED_PORT_SHUTDOWN
shutdown
!
interface FastEthernet0/6
description UNUSED_PORT_SHUTDOWN
shutdown
!
interface FastEthernet0/7
description UNUSED_PORT_SHUTDOWN
shutdown
!
interface FastEthernet0/8
description UNUSED_PORT_SHUTDOWN
shutdown
!
```

```
interface FastEthernet0/9
description UNUSED_PORT_SHUTDOWN
shutdown
!
interface FastEthernet0/10
description UNUSED_PORT_SHUTDOWN
shutdown
!
interface FastEthernet0/11
description UNUSED_PORT_SHUTDOWN
shutdown
!
interface FastEthernet0/12
description UNUSED_PORT_SHUTDOWN
shutdown
!
interface FastEthernet0/13
description UNUSED_PORT_SHUTDOWN
shutdown
!
interface FastEthernet0/14
description UNUSED_PORT_SHUTDOWN
shutdown
!
interface FastEthernet0/15
description UNUSED_PORT_SHUTDOWN
shutdown
!
interface FastEthernet0/16
description UNUSED_PORT_SHUTDOWN
shutdown
!
interface FastEthernet0/17
```

```
description UNUSED_PORT_SHUTDOWN
shutdown
!
interface FastEthernet0/18
description UNUSED_PORT_SHUTDOWN
shutdown
!
interface FastEthernet0/19
description UNUSED_PORT_SHUTDOWN
shutdown
!
interface FastEthernet0/20
description UNUSED_PORT_SHUTDOWN
shutdown
!
interface FastEthernet0/21
description UNUSED_PORT_SHUTDOWN
shutdown
!
interface FastEthernet0/22
description UNUSED_PORT_SHUTDOWN
shutdown
!
interface FastEthernet0/23
description UNUSED_PORT_SHUTDOWN
shutdown
!
interface FastEthernet0/24
description UNUSED_PORT_SHUTDOWN
shutdown
!
interface GigabitEthernet0/1
description UPLINK_TO_CORE_1
```

```
switchport mode trunk
!
interface GigabitEthernet0/2
description UPLINK_TO_CORE_2
switchport mode trunk
!
interface Vlan1
no ip address
shutdown
!
interface Vlan99
ip address 192.168.99.10 255.255.255.0
!
ip default-gateway 192.168.99.1
!
banner motd ^C
*****
* WARNING: AUTHORIZED ACCESS ONLY. *
* ALL ACTIVITIES MONITORED. *
*****
^C
!
!
!
line con 0
password 7 0822455D0A16
logging synchronous
login
!
line vty 0 4
password 7 0822455D0A16
login
transport input ssh
```

```
line vty 5 15
password 7 0822455D0A16
login
transport input ssh
!
!
!
!
end
```

```
Access-1#
```